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Unified Circular Economy Platform for E-Waste Recovery and Product Repair Incentivization

Repair-first decisions • Smart e-waste recovery • Reward-driven sustainability



TEAM MEMBERS

Asmitha Kadiri(192425038)
Madhu Sneha Latha Reddy(192425092)
Pragna(192311150)



PROJECT GUIDE

Dr. SivaSankar



SUBJECT DETAILS

CSA-0925
Java Programming

Problem Background

Where this project sits, and why it's needed now



Application Domain

Consumer electronics lifecycle management — bridging device repair services and e-waste recycling within a single sustainability-focused platform.



Current Scenario

Repair shops and recycling centers operate as disconnected, local, often informal services with no shared digital infrastructure.



Need for the Project

Users lack a simple, guided way to decide between repairing a device or recycling it responsibly — leading to premature disposal.



Target Users

Everyday device owners, certified repair vendors, e-waste recyclers, and institutional administrators overseeing the ecosystem.

Problem Statement

The scale of the issue, and why current systems fall short

E-waste is one of the fastest-growing waste streams worldwide, while millions of repairable devices are discarded prematurely — largely because users have no easy way to compare repair against recycling.

LIMITATIONS OF EXISTING SYSTEMS

- ⚠️ Repair and recycling services run on separate, disconnected platforms
- ⚠️ No structured way to weigh repair cost against recycling benefit
- ⚠️ Zero incentive for choosing responsible disposal or recycling
- ⚠️ Decisions rely entirely on manual research and guesswork

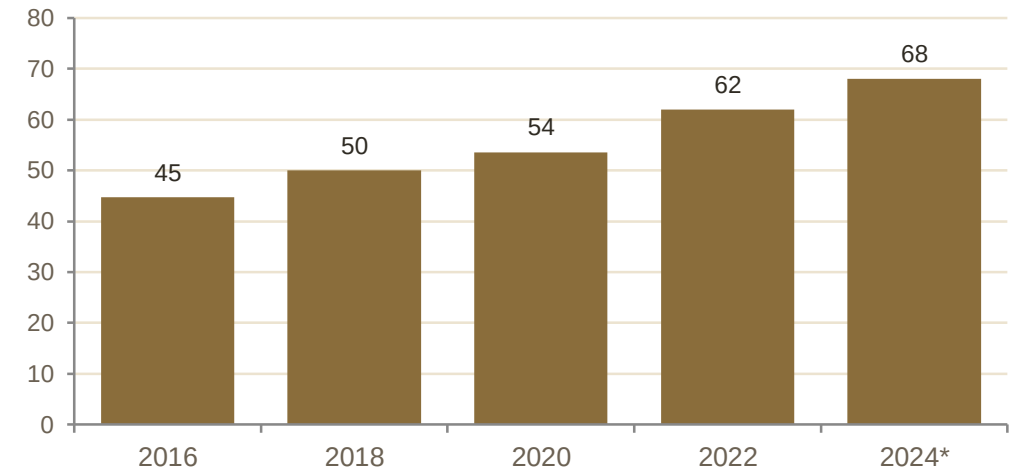
62M+

Tonnes of global
e-waste generated (2022)

<25%

Share formally
collected & recycled

GLOBAL E-WASTE GENERATED (Mt)



**Projected. Source: Global E-waste Monitor*

Objectives

What the platform is designed to achieve



PRIMARY OBJECTIVE

Build a unified Java-based platform that guides repair-first decisions and rewards responsible e-waste recycling.

01



Repair vs Recycle

Help users determine whether a device should be repaired or recycled.

02



Connect to Vendors

Connect users with nearby, verified repair service providers.

03



Reward Recycling

Reward responsible e-waste recycling through redeemable points.

04



Circular Economy

Promote sustainable practices through a circular economy approach.

Project Scope: A working web-desktop application (Java Swing + JDBC + MySQL) covering device submission, repair booking, vendor workflow, recycling scheduling, and a reward engine — for a single-region pilot deployment.

Literature Survey

Recent research (2021–2026) informing this project

Author(s)	Year	Method	Limitation
Islam, Jony, Hasan et al.	2023	Two-stream image transformer (EwasteNet) for e-waste image classification	Classifies waste only — no repair recommendation or incentive layer
SAFA-E system authors	2022	Android + web app for e-waste submission and disposal guidance	Covers recycling disposal only; repair is not considered as an option
Sustainable computing review	2024	Conceptual sustainable-computing framework for IT e-waste	High-level review; no deployable, user-facing decision system
Incentivization scheme authors	2023	Vector-space incentive scoring with smart contracts for collection	Reward logic isolated from repair workflow; blockchain adds overhead
Alnuaimi, Alsafi, Alshehhi, Zemerly	2023	Ethereum blockchain + decentralized storage for recyclable tracking	Targets plastic waste, not electronics; no device assessment step
End-to-end collection & sorting authors	2024	QR-code based collection portal with ML-assisted sorting	Backend logistics focus; no consumer-facing repair or reward module
Integrated e-waste management review	2023	Comprehensive review of collection, treatment and mitigation stages	Survey-level; does not propose an implementable software platform

Existing vs Proposed System

What changes with a unified, incentive-driven platform

EXISTING SYSTEM

-  Separate, unconnected repair platforms
-  Separate e-waste recycling services
-  No reward or incentive system
-  Manual, guesswork-based decision making

Fragmented, manual

PROPOSED SYSTEM

-  One unified platform for repair + recycling
-  Smart repair-vs-recycle recommendation engine
-  Built-in reward points for responsible recycling
-  Repair-first approach, better sustainability outcomes

Novelty: Unified + Smart + Rewarding

Requirements

What the system needs, functionally and technically



Functional

- User registration & login
- Device submission & assessment
- Repair booking & tracking
- Recycling scheduling & rewards



Non-Functional

- Responsive, intuitive UI
- Data security & role-based access
- Reliable, low-latency performance
- Scalable to more regions/vendors



Hardware

- Intel i3 (or higher) processor
- 4 GB RAM minimum
- 250 GB storage
- Standard internet connectivity



Software

- Java Swing (frontend)
- Java (backend logic)
- NetBeans / IntelliJ / Eclipse
- Windows / Linux OS



Database

- MySQL relational database
- JDBC connectivity layer
- 8 structured core tables
- Regular backup routine

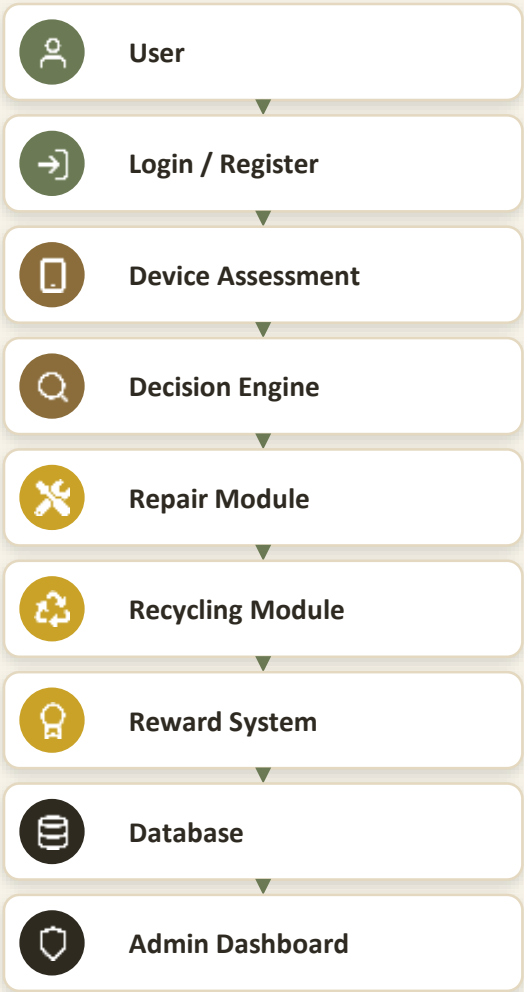


Technology Stack

- Frontend: Java Swing
- Backend: Java
- DB: MySQL via JDBC
- IDE: NetBeans / IntelliJ / Eclipse

System Architecture

How a request flows through the platform



Device Assessment

Captures device details and condition to feed the recommendation logic.

Decision Engine

Weights repair cost against recycling value to suggest the best path.

Repair Module

Books nearby vendors, tracks status, and stores repair history.

Recycling & Rewards

Schedules pickup and credits reward points for redemption.

Database & Admin

MySQL via JDBC persists all records; admins monitor the full system.

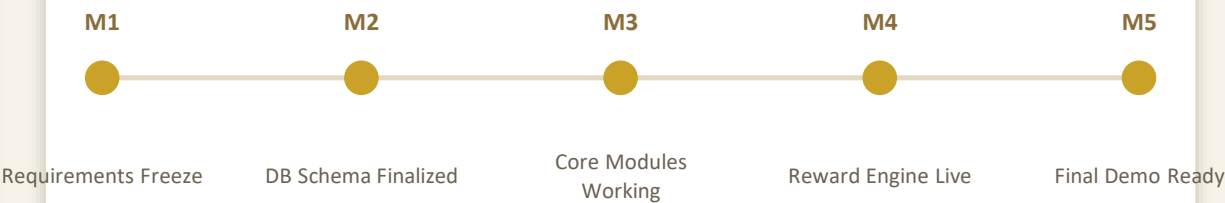
Project Planning

Breakdown, schedule, milestones, and risk outlook

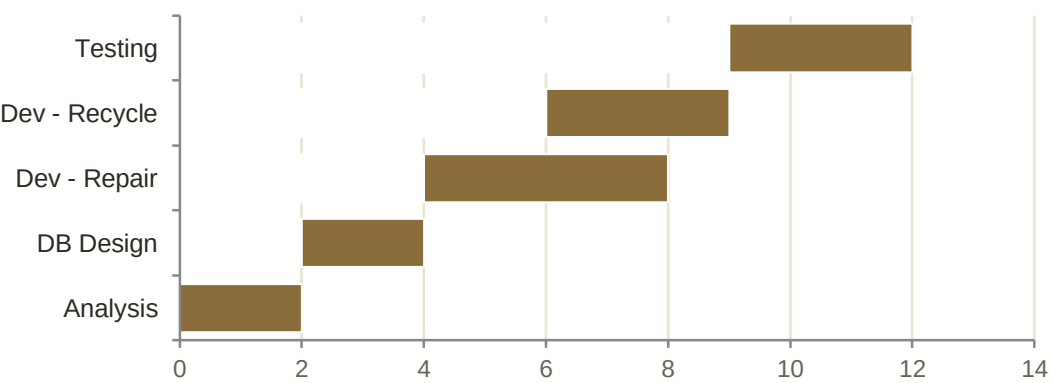
WORK BREAKDOWN STRUCTURE

- 1 Requirement Analysis & Design
- 2 Database & Backend Development
- 3 Repair & Recycling Modules
- 4 Reward Engine + Admin Panel
- 5 Testing, Integration & Deployment

MILESTONES



GANTT CHART (WEEKS)



RISK ANALYSIS

Risk	Impact	Mitigation
Vendor data delays	Medium	Early vendor onboarding & manual seed data
Scope creep	Medium	Fixed module boundaries, phased delivery
DB integration issues	Low	Early JDBC integration testing
Reward abuse/fraud	Medium	Point caps + admin verification step

Expected Outcomes



Deliverables

Working Java platform with repair booking, recycling scheduling, and reward redemption.



Benefits

Faster repair-or-recycle decisions and one login for the whole e-waste journey.



Environmental Impact

Longer device lifespans and higher, better-tracked recycling participation.



Future Scope

AI recommendations, QR tracking, Maps integration, carbon calculator, mobile app.

REFERENCES

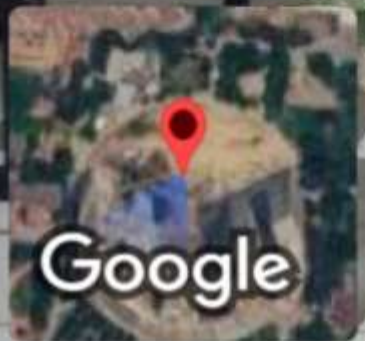
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Thank You



GPS Map Camera



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22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India

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